

Bonus Project Exposé

Deep Learning – SoSe 2026

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Group-23

1. Additional Dataset

To evaluate generalization, we propose the “Crypto Data Hourly Price” dataset Julien (2023), comprising over 1.12 million rows of hourly Open, High, Low, Close, Volume (OHLCV) and trade count data across 34 major cryptocurrencies from August 2017 to October 2023. Our objective is to forecast hourly trading volume across these highly correlated assets. This dataset provides a rigorous stress test for our architecture.

2. Model Architecture

We plan to use TiDE (Das et al., 2023), an MLP-based encoder–decoder designed for long-horizon multivariate forecasting with known future covariates. We chose it because the benchmark dataset provides rich future-dated covariates (the *_forecast and risk columns), which TiDE can directly incorporate, and because its dense-MLP design trains quickly and produces the full 336-step horizon in one shot, avoiding the error accumulation of autoregressive rollout.

3. Planned Improvements or Variants

Our planned work targets the core structural challenges of this benchmark: executing a long-horizon 336-step forecast, managing scale heterogeneity, and handling partially missing exogenous covariates.

- **Improvement 1 – Uncertainty Quantification via Conformal Prediction:** Point forecasts inherently degrade in reliability over extended horizons. To address this, we will implement a split conformal prediction framework to generate statistically valid prediction intervals alongside our point estimates (Angelopoulos & Bates, 2021). By calibrating non-conformity scores (e.g., scaled absolute residuals) on the held-out validation set, we aim to provide distribution-free coverage for the 336-step forecast, proving robustness under scale heterogeneity.
- **Improvement 2 – Missingness-Robust Covariates:** To handle the 4.5% missing covariate data without destabilizing the decoder, we will combine hour-of-week median imputation with binary missingness indicators (Che et al., 2018). We will also apply co-variate dropout during training to ensure the network’s predictive accuracy degrades gracefully rather than catastrophically under imperfect input. Furthermore we will also conduct a feature-group ablation study grouping the co-variables to isolate feature importance.
- **Improvement 3 – Scale Stabilization & Objectives:** We will apply Reversible Instance Normalization (RevIN) (Kim et al., 2022) to manage target scale variance. To ensure we beat the seasonal baseline, we will explicitly deseasonalize the targets by subtracting historical hour-of-week profiles during preprocessing, forcing the network to exclusively predict residual deviations. Finally, because the leaderboard metric (WAPE, MAE) heavily penalize L1 errors, we will align our training objective by optimizing with a Huber/MAE loss rather than standard MSE.

4. Baselines for Comparison

We will evaluate our model against the naive last-value baseline, which propagates the final observed historical value across the entire horizon to test for basic signal extraction. Additionally, the seasonal mean baseline will be used to predict historical hour-of-week averages, ensuring our architecture can map stochastic residuals rather than simply memorizing deterministic periodic cycles. Finally, we propose DLinear as our primary structural deep learning baseline Zeng et al. (2023). DLinear is a highly efficient, channel-independent model that uses moving average decomposition and simple linear layers to consistently outperform complex Transformers in long-horizon tasks. Comparing our dense MLP architecture against DLinear will rigorously validate the advantage of cross-variate feature mixing over strictly univariate temporal mappings.

5. Risks and Open Questions

- **Risk 1 – TiDE Generalization under Missing Covariates and Heterogeneity:** TiDE was originally evaluated on standard long-horizon benchmarks (Das et al., 2023); its performance on a dataset with 4.5% missing covariates and high cross-series heterogeneity (96 series with different `nominal_capacity` ranges) is uncertain. While our missingness-robust covariate strategy (Che et al., 2018) mitigates part of this risk, catastrophic performance degradation under high missingness rates at inference time remains an open concern.
- **Risk 2 – Conformal Prediction Calibration over Long Horizons:** Our conformal prediction framework requires a well-calibrated residual distribution on the validation set (Angelopoulos & Bates, 2021). If the residuals are heavily non-stationary or heavy-tailed across the 336-step horizon, the coverage guarantees may not hold in practice.
- **Open Question 1 – Optimal Input Context Length:** We have not yet determined the optimal input context length for TiDE (Das et al., 2023) on this dataset. A context window that is too short may miss long-range dependencies, while one that is too long increases computational cost and risks overfitting to noisy historical patterns. This will be addressed through validation-set experiments.
- **Open Question 2 – Transferability to the Additional Dataset:** The transferability of our TiDE architecture (Das et al., 2023) to the crypto dataset (Section 1) is an open question, particularly since the additional dataset lacks the future-dated covariates present in the benchmark, requiring TiDE to operate in a covariate-free regime. This will be explored empirically.

References

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